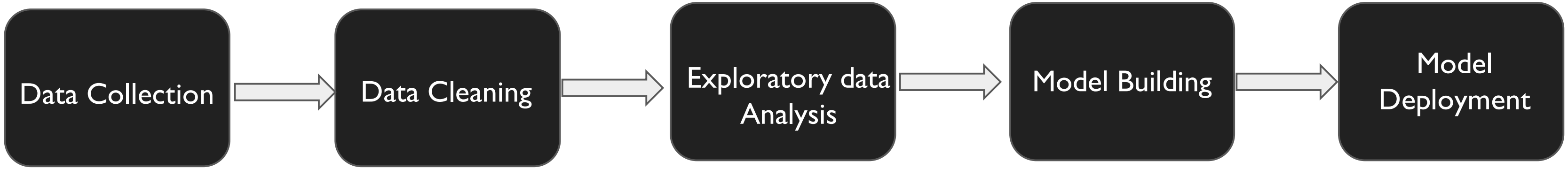
Data Preparation

Contents

Data Science Life Cycle



Data Collection

- Data Collection is the initial step in any data-driven project
- It involves gathering raw data from various sources
- Quality data collection sets the foundation for accurate analysis and decision-making
- Data Collection can be challenging due to issues like missing data, inconsistencies, and noise

Data Cleaning

- Data Cleaning is the process of identifying and correcting errors, inconsistencies and inaccuracies in the collected data.
- It involves tasks like handling missing values, removing duplicates, and correcting wrong entries.

Steps for Data Cleaning

Handling Missing Data:

- Decide on strategies like imputation (mean, median, mode)

Dealing with Duplicates:

- Identify and remove duplicate entries

Standardizing Formats:

- Ensure consistency in data formats, units, and scales for accurate analysis.

Exploratory Data Analysis

- EDA is a crucial step before in-depth analysis.
- It involves visually and statistically summarizing the main characteristics of the data

Exploratory Data Analysis

Understanding Data Distribution:

- Histograms, box plots, and density plots help identify data ranges and anomalies.

Identifying Relationships:

- Scatter plots and correlation matrices reveal connections between variables.

Detecting Outliers:

- Visualization and statistical methods help in spotting extreme values.

Introduction to Problem Statement

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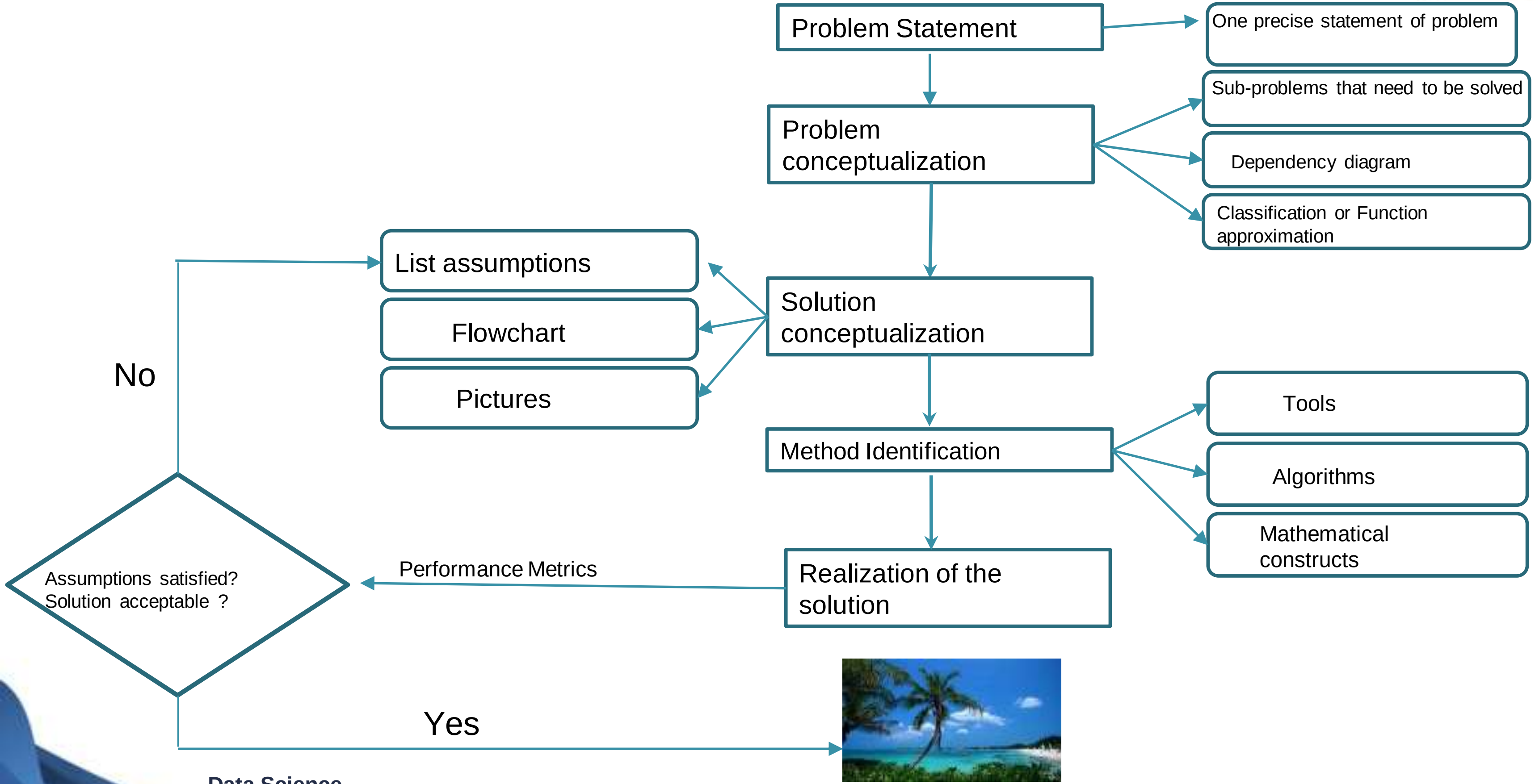
- A problem statement is a clear and concise description of a specific issue, challenge, or opportunity that needs to be addressed.

- It defines the problem in a way that allows others to understand its nature and scope.

Conceptual Framework for Solving Data Analysis Problems



CASE STUDY

Sources to collect Cricket Data:

Websites:

- Kaggle - https://www.kaggle.com/datasets/
- ESPN cricinfo - https://stats.espncricinfo.com/ci/engine/stats/index.html
- Howstat - http://www.howstat.com/cricket/statistics/players/playermenu.asp
- GitHub Repositories - https://github.com/topics/cricket-data

Sources to collect Cricket Data:

API:

Many sports leagues offer APIs that developers can use to access real-time and historical data. These APIs provide structured data that can be used for various analytical purposes.

- Sportmonks
- Statscore

Sensor Technology:

Wearable sensors are used to collect player movement and performance data.

Problem Statement

Comprehensive analysis on the batting performance of Virat Kohli in IPL 2022. Analyzing various aspects of his performance, such as runs scored, strike rate at different phases,boundaries hit (4s and 6s).

Objective:

Identify his strengths and weaknesses, and provide insights on his overall performance.

Approach

Solution Conceptualization:

- Identify if data is clean
- Look for missing data and duplicates.
- Identify variables to perform data analysis to calculate metrics that evaluates performance of a batsman.
- Identify the strengths and weaknesses by analyzing metrics that evaluates performance of a batsman.

Variable Description

Size: 306 x 17

Data file: virat_kohli_ball_by_ball_2022.csv

Variables	Data type	Description
ID	Integer	An identifier for each row.
innings	Integer	Indicates the innings number in which the delivery was played (1st or 2nd innings).
overs	Integer	Represents the over number in which the delivery was bowled.
ballnumber	Integer	The specific ball number of the over
batter	String	Name of the batsman (Virat Kohli).

Variable Description

Size: 306 x 17

Data file: virat_kohli_ball_by_ball_2022.csv

Variables	Data type	Description
bowler	String	Name of the bowler who bowled the delivery.
non-striker	String	Name of the non-striker at the crease during the delivery.
extra_type	String	Type of extras scored in that delivery (wide, no-ball, etc.).
batsman_run	Integer	Number of runs scored by Virat Kohli on that delivery.
extras_run	Integer	Number of extra runs scored on that delivery.

Variable Description

Size: 306 x 17

Data file: virat_kohli_ball_by_ball_2022.csv

Variables	Data type	Description
total_run	Integer	Total runs scored on that delivery (including batsman runs and extras).
non_boundary	Integer	Indicates if the runs scored on the delivery were from non-boundary hits.
isWicketDelivery	Integer	A binary indicator (1 or 0) representing whether Virat Kohli got out on that delivery or not.
player_out	String	Name of the player who got out (if isWicketDelivery is 1).
kind	String	Type of dismissal (e.g caught, bowled, run-out, etc.).

Variable Description

Size: 306 x 17

Data file: virat_kohli_ball_by_ball_2022.csv

Variables	Data type	Description
fielders_involved	String	Names of fielders involved in the dismissal.
BattingTeam	String	Name of the team for which Virat Kohli was batting.

Dataset source : *https://github.com/samar4saeedkhan/IPL-2022-Analysis/blob/main/DATA/IPL_Ball_by_Ball_2022.csv*

Data Collection

Load the virat_kohli_ball_by_ball_2022.csv csv file using

read_csv() function

```
# Load the dataset 'virat_kohli_ball_by_ball_2022.csv'
data = pd.read_csv('/content/virat_kohli_ball_by_ball_2022.csv')

data.head()
```

Data Collection

Load the virat_kohli_ball_by_ball_2022.csv csv file using read_csv( ) function

	ID	innings	overs	ballnumber	batter	bowler	non-striker	extra_type	batsman_run	extras_run	total_run	non_boundary	isWicketDelivery	player_out	kind	fielders_involved	BattingTeam
0	1304049	1	7	2	V Kohli	Harpreet Brar	F du Plessis	NaN	1	0	1	0	0	NaN	NaN	NaN	Royal Challengers Bangalore
1	1304049	1	7	4	V Kohli	Harpreet Brar	F du Plessis	NaN	0	0	0	0	0	NaN	NaN	NaN	Royal Challengers Bangalore
2	1304049	1	7	5	V Kohli	Harpreet Brar	F du Plessis	NaN	1	0	1	0	0	NaN	NaN	NaN	Royal Challengers Bangalore
3	1304049	1	8	1	V Kohli	RD Chahar	F du Plessis	NaN	1	0	1	0	0	NaN	NaN	NaN	Royal Challengers Bangalore
4	1304049	1	9	1	V Kohli	Harpreet Brar	F du Plessis	NaN	2	0	2	0	0	NaN	NaN	NaN	Royal Challengers Bangalore

Data Cleaning

Check for any duplicate rows and missing values.

Duplicates

```
[ ] duplicate = data[data.duplicated()]

    print("Duplicate Rows :")

    duplicate
```

Duplicate Rows :

ID	innings	overs	ballnumber	batter	bowler	non-striker	extra_type	batsman_run	extras_run	total_run	non_boundary	isWicketDelivery	player_out	kind	fielders_involved	BattingTeam

Data Cleaning

Check for any duplicate rows and missing values.

Missing values

```
[ ]  data.isna().sum()
```

```
ID                    0
innings               0
overs                 0
ballnumber            0
batter                0
bowler                0
non-striker           0
extra_type          292
batsman_run           0
extras_run            0
total_run             0
non_boundary          0
isWicketDelivery      0
player_out          290
kind                290
fielders_involved   293
BattingTeam           0
dtype: int64
```

Data Analysis

We calculate mainly total runs, strike rate, batting average and batting performance at different phases to evaluate the batsman performance.

Three phases:

- Powerplay (0 - 6)
- Middle overs (7 - 15)
- Death overs (16 - 20)

Data Analysis

Total Runs:

To calculate the total runs we sum the batsman_run column.

```
# Calculate the total runs scored by Virat Kohli in IPL 2022
total_runs = data['batsman_run'].sum()

print("Total Runs:",total_runs)
```

```
Total Runs: 341
```

Data Analysis

Strike Rate of a batsman:

$$\text{Strike Rate} = \left[\frac{\text{Total runs made by batsman}}{\text{Total balls faced by batsman}} \right] \text{x } 100$$

Data Analysis

Strike Rate of a batsman:

```
[ ]  total_runs = data['batsman_run'].sum()

     strike_rate = (total_runs/total_balls_faced) * 100

     print(f"Strike rate:{strike_rate:.2f}")

     Strike rate:115.99
```

Data Analysis

Batting Average:

$$\text{Batting Average} = \frac{\text{Total runs made by batsman}}{\text{Number of times batsman got out}}$$

```
[ ]  # How many times Virat got out
     times_out = data[data['player_out'] == 'V Kohli']['isWicketDelivery'].sum()

     print("Number of times Virat Kohli got out:", times_out)
```

```
Number of times Virat Kohli got out: 14
```

```
[ ]  batting_avg = total_runs/times_out

     print(f"Batting Average of Virat Kohli:{batting_avg:.2f}")
```

```
Batting Average of Virat Kohli:24.36
```

Data Analysis

Batting performance at different phases:

	Total Runs	Kohli Runs	Balls Faced	Strike Rate
Powerplay	181	168	158	106.32
Middle overs	159	151	129	117.05
Death overs	22	22	19	115.78

Conclusion

- Virat Kohli scored a total of 341(294) runs in IPL 2022.
- Virat Kohli maintained strike rate of 117.59.
- Virat Kohli maintained batting average of 24.56 in IPL 2022.
- He managed to maintain a decent overall strike rate in the middle overs and contributed significantly with 40 boundaries to his total of 341 runs in IPL 2022.

Conclusion

- One major issue for Kohli is low power play strike rate. Instead of finding the right balance between singles and boundaries, Kohli settled for more singles.

- Overall, it was a challenging IPL season for Virat Kohli.

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THANK YOU